Final paper

Are Catastrophe Bonds Really Zero-Beta Assets?

Diversification Benefits in Normal Markets and Financial Crises

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1. Introduction

A catastrophe (CAT) bond lets an insurer, a reinsurer or a government pass part of its exposure to natural disasters to capital-market investors. The sponsor pays a premium, the investors put up principal that is held as collateral, and if a specified hurricane, earthquake or other event occurs before maturity, part or all of that principal goes to the sponsor (Cummins, 2008). Securitisation opened catastrophe risk to investors with far more capital than reinsurers, and the market grew accordingly: outstanding CAT bond risk capital rose from about $2.2 billion in 2002 to $23 billion in June 2017 (Drobetz et al., 2020).

To attract those investors, the market offered a simple argument. Whether a hurricane hits Florida or an earthquake hits Tokyo does not depend on the stock market, so the losses on a CAT bond should be unrelated to the losses on stocks and corporate bonds. In CAPM terms, CAT bonds would be zero-beta assets, and any spread above expected losses would reward a risk that adds nothing to the systematic risk of a diversified portfolio. But the investor holds a financial security, not the natural event: a contract with a special purpose vehicle, whose principal sits in a collateral account, whose price comes from dealer quotes, and which is held by funds that may need cash when everyone else does. Each of these features can carry financial stress into the bond when no catastrophe has happened, and the published evidence on whether they do is contradictory.

Three positions can be distinguished. Haffar and Le Fur (2022) come closest to the zero-beta view, although they do not test it formally: in their sample, CAT bonds add the least to extreme joint movements with an equity portfolio. Carayannopoulos and Perez (2015) and Gürtler, Hibbeln and Winkelvos (2016) reject it: CAT bond returns became correlated with the stock market after the collapse of Lehman Brothers, and CAT bond premiums move with corporate bond spreads. Drobetz, Schröder and Tegtmeier (2020) also conclude that CAT bonds do not behave like a zero-beta investment. They find, however, that CAT bonds acted as a safe haven against extreme stock market declines after the financial crisis, a result that Feng and Zeng (2025) extend to the COVID-19 period. In this third position the answer depends on the period.

The difference matters for investors and for risk managers. A diversifier lowers the variance of a portfolio on average; a safe haven does not move with the rest of the portfolio in a crash, so an investor who buys CAT bonds for the second property when they only have the first is exposed at the worst moment, and a value-at-risk estimated on calm data understates the risk if the correlation rises when stocks fall.

The three articles cannot settle the question as they stand, because they do not measure the same object. They use different series, frequencies, windows and definitions of stress (Table 1). None separates a crisis in which the collateral is at risk from one in which investors only need cash. Their conclusions contradict each other; their estimates need not.

My research question is: **to what extent do CAT bonds retain their diversification benefits during financial stress, and why do published studies reach different answers?** I argue that the disagreement has little to do with whether catastrophes occur and much to do with how

catastrophe risk is collateralised, priced and measured. My contribution has three parts. The first is a decomposition of the CAT bond return into an insurance core, nearly independent of markets for natural perils, and a financial wrapper, whose market exposure depends on the collateral structure and on the type of stress; it sorts the three samples into four regimes. The second is a re-reading of the safe-haven evidence: the coefficients of Drobetz et al. imply a higher correlation in the worst 1 percent of post-crisis weeks than in ordinary weeks. The third is a prediction none of the articles makes: the safe-haven property depends on who holds the bonds. The answer it gives is conditional: CAT bonds remain a reliable diversifier, but their safe-haven property requires safe collateral and weakens when holders are forced to sell.

The question raises three issues: a conceptual one, since diversifier, hedge, safe haven and zero beta are different properties; an economic one, since a CAT bond can move with the market because its collateral is exposed to a bank or because its holders must sell, two causes with different remedies; and a statistical one, since betas, correlations and copulas can disagree on the same bonds.

The paper proceeds as follows. Section 2 explains how CAT bonds work, defines the properties tested, presents the three articles and related studies, and states what the literature leaves open. Section 3 assesses the articles, presents my contribution, the decomposition of the CAT bond return, applies it to four stress regimes, discusses measurement and rival explanations, proposes tests and portfolio rules, and assesses the validity of the analysis. Section 4 concludes.

2. Literature review

The three articles were selected on three criteria: they study the same question, whether CAT bonds keep their diversification benefit under stress; they reach divergent conclusions; and together they cover the three designs used in this literature, regime-switching correlations on a crisis window, quantile-dummy regressions on dynamic correlations, and copula dependence on a calm window. Other studies enter only when they test the same property on another period (Feng and Zeng, 2025) or document a mechanism that the articles invoke (Gürtler et al., 2016; Braun, 2016; Herrmann and Hibbeln, 2023).

2.1 How a CAT bond works, and where financial risk can enter

A sponsor sets up a special purpose vehicle (SPV) that sells notes to investors and signs a reinsurance contract with the sponsor. The proceeds of the notes are placed in a collateral trust account. Investors receive a floating coupon, a money-market rate plus a risk premium, and the bonds usually mature in three to four years (Drobetz et al., 2020). If a qualifying event occurs, the collateral is released to the sponsor and investors lose part or all of their principal. The trigger can be the sponsor’s own losses, an industry loss index, a modelled loss or a physical parameter such as wind speed (Cummins, 2008).

Several parts of this structure have nothing to do with nature. One is the **collateral**. Before 2009, the vast majority of CAT bonds used a total return swap: an investment bank converted the return on the collateral into a money-market rate and was supposed to absorb changes in the value of the

collateral. The collateral itself often had much longer maturities than the bond and could include highly rated asset-backed and mortgage-backed securities, with little monitoring and no obligation for the swap counterparty to keep its value above a threshold (Carayannopoulos and Perez, 2015). When Lehman Brothers, the swap counterparty of several CAT bonds, failed in September 2008, this structure failed with it. After issuance froze in late 2008, new deals restricted collateral to cash, government-guaranteed bank debt, U.S. Treasury money-market funds and putable notes from sovereign or quasi-governmental issuers, with frequent mark-to-market and trustee monitoring (Carayannopoulos and Perez, 2015).

Another is **liquidity**. CAT bonds trade over the counter and not very often. Using 9,393 secondary-market trades reported to TRACE between 2015 and 2019, Herrmann and Hibbeln (2023) find that liquidity is lower for riskier bonds and for bonds far from maturity, and they estimate that the liquidity premium accounts for about a fifth of the observed CAT bond yield spread: 98 basis points on average and 141 basis points for high-risk bonds. Liquidity is also higher when the whole market trades more, and trading is dominated by brokers who hold no inventory of their own, so few intermediaries stand ready to absorb sales. A premium of that size can move when dealers and investors run short of capital, which is also when stock prices fall (Brunnermeier and Pedersen, 2009). In late 2008, distressed CAT bonds were put up for sale quickly in a market drained of liquidity, and investors suffered losses "not from natural catastrophic events, but from market risks that they thought they had been isolated from" (Carayannopoulos and Perez, 2015, pp. 22-23).

A third is **measurement**. The three articles use two Swiss Re indices, which are built from secondary-market bid indications rather than trades (Drobetz et al., 2020), and the net asset value of a CAT bond fund. An index of quotes can be smoothed, which lowers its measured volatility and its correlation with other markets (Getmansky, Lo and Makarov, 2004).

2.2 Diversifier, hedge and safe haven: what each test measures

The articles use, explicitly or not, the definitions of Baur and Lucey (2010). A hedge is uncorrelated or negatively correlated with another asset or portfolio on average. A diversifier is positively, but not perfectly, correlated with it on average. A safe haven is uncorrelated or negatively correlated with it during extreme market declines. Baur and McDermott (2010) add that a strong hedge or safe haven is negatively correlated with the market and a weak one is uncorrelated. The zero-beta claim, which says the beta is zero at all times, therefore amounts to a weak hedge plus a weak safe haven, and evidence of diversification does not confirm it. The definitions are also about correlation, not price: a statistical safe haven can lose money in a crash as long as its losses do not move with the market's.

Two warnings follow. The word "strong" changes meaning between studies: in the regression test of Drobetz et al. (2020), a strong safe haven is an asset whose correlation falls significantly in extreme weeks, not one whose correlation is negative. And betas, correlations and copulas measure dependence differently, so they can disagree on the same data (Section 3.4).

2.3 Carayannopoulos and Perez (2015): zero beta, except in a crisis

The first article asks whether CAT bonds are zero-beta investments and whether they diversify a portfolio. The authors use 614 weekly returns, from January 2002 to October 2013, on the Swiss Re BB CAT bond price index, the S&P 500, an investment-grade U.S. corporate bond index and a 5-to-7-year government bond index. They use a price index because the S&P 500 is one, and their results hold with the broad Swiss Re index and with the BB total return index. They estimate time-varying covariances with a diagonal-VEC multivariate GARCH model, identify high- and low-correlation regimes with a Markov-switching model, and compute dynamic hedge ratios, the conditional covariance with the S&P 500 divided by its conditional variance, which they read as time-varying betas. The crisis is the NBER period from December 2007 to May 2009.

Simple correlations already show the pattern. The correlation between CAT bonds and the S&P 500 is 0.016 before the crisis, 0.252 during it (significant at 5 percent) and 0.058 after it; with corporate bonds it goes from 0.003 to 0.475 and back to 0.092. With government bonds, by contrast, it falls from 0.128 to an insignificant 0.054: in the crisis, CAT bonds moved with credit, not with safe fixed income. The dynamic estimates add the timing. The correlation with the stock market stayed close to zero until the end of August 2008, rose from September 2008 to 0.24 in May 2009, fluctuated widely through 2010, and returned to its pre-crisis level at the beginning of 2011. The economic size of the effect was small. The CAT bond beta peaked at 0.014 in January 2010, against 0.13 for corporate bonds, and the weekly volatility of CAT bonds barely moved (0.37 percent before the crisis, 0.41 percent during it) while that of the S&P 500 rose from 1.91 to 4.49 percent.

The regime estimates agree. The high-correlation regime with stocks that began in October 2008 lasted about 30 months; those after Katrina, the 2011 Japanese earthquake and Sandy were much shorter. The regime with corporate bonds was less persistent, and the correlation with government bonds shows no crisis regime at all.

The authors conclude that CAT bonds “are not zero-beta assets” (p. 21), that weaknesses of the trust accounts were the main cause, and that CAT bonds remain a valuable diversifier because the crisis affected them far less than other assets. They also note that Hurricane Ike struck Texas in the same month as Lehman’s failure and argue, by comparison with the other events, that Ike acted as a catalyst but cannot explain the size and persistence of the 2008 correlation. They describe the effect of the crisis as temporary, since the correlations return to their pre-crisis levels after it.

2.4 Drobetz, Schröder and Tegtmeier (2020): a diversifier, and a safe haven only after the crisis

The second article widens the comparison to an international multi-asset portfolio. The authors pair the Swiss Re Global Unhedged CAT Bond Performance Index with total return indices for world stocks (MSCI World IMI), global government bonds, listed real estate, commodities, listed private equity and infrastructure. They use 885 weekly returns from January 2002 to December

2018, split into a pre-crisis period (2002-2008) and a post-crisis period (2009-2018). They first estimate dynamic conditional correlations (DCC), a weekly series of correlations between CAT bonds and each asset class, then regress them on dummies for the weeks in which the other asset's return falls below its 10, 5 and 1 percent quantiles: $DCC_t = m_0 + m_1 \cdot d_{10\%} + m_2 \cdot d_{5\%} + m_3 \cdot d_{1\%}$. A significantly positive m_0 makes CAT bonds a diversifier, a zero (negative) m_0 a weak (strong) hedge. Insignificant dummy coefficients make them a weak safe haven, significantly negative ones a strong safe haven.

The constant m_0 is positive and significant at 1 percent for every asset class: about 0.11 with stocks over the full period, 0.24 before 2009 and 0.10 after. CAT bonds are therefore a diversifier against all six asset classes and a hedge against none. Before 2009 several dummy coefficients are positive, including 0.072 for the 5 percent stock quantile, so the correlation rose in extreme weeks. After 2009 only one coefficient is negative and significant: m_2 for stocks, −0.091 at the 10 percent level, or −0.101 at the 5 percent level with desmoothed returns. On this coefficient rests the conclusion that CAT bonds are a strong safe haven against extreme stock market declines only in the post-crisis period; they are a safe haven against none of the other asset classes. The authors conclude that CAT bonds are "not immune to changes in systematic risk" and that their behaviour "is not consistent with a zero-beta investment" (p. 5), and they relate this to the two mechanisms already documented: the weak trust accounts described by Carayannopoulos and Perez (2015) and the dependence of CAT bond premiums on corporate spreads found by Gürtler et al. (2016).

The summary statistics show the two faces of the asset. The CAT bond index has the lowest weekly volatility of the seven series (0.76 percent) but the highest kurtosis (313.5) and a strongly negative skewness (−9.0), the mark of rare but large event losses. Its first-order autocorrelation is significantly negative (−0.34), which the authors attribute to the indicative pricing of the index; they re-estimate every regression on desmoothed returns and find the estimates "virtually unchanged" (p. 5).

2.5 Haffar and Le Fur (2022): little extreme co-movement

The third article studies the dependence structure. Its CAT bond series is not an index but a CAT bond fund, the GAMSTAR Institutional USD Acc share class managed by GAM, whose monthly returns are compared with six stock indices (CAC 40, DJIA, EURO STOXX 50, FTSE 100, Hang Seng, Nikkei 225) over 100 months, from November 2011 to February 2020. The fund holds 92 percent fixed income and 8 percent cash and is benchmarked against euro government bonds; the paper dates its launch to 26 March 2012. The authors model each series with an ARMA-GARCH process and join them with a Student-t copula with asymmetric dynamic conditional correlation. They then compare risk-adjusted performance, contributions to the portfolio's value-at-risk and expected shortfall, co-moments with an equally weighted benchmark portfolio of the other assets, and mean-variance optimal portfolios.

Their findings are the most favourable of the three. The fund has the highest modified Sharpe ratio (0.112, against 0.027 to 0.089 for the stock indices). Its beta with the benchmark portfolio is

0.031, against 0.77 to 1.27 for the stock indices, and its "beta co-kurtosis" is 0.015, against 0.80 to 1.35. Its contribution to the portfolio's 99 percent value-at-risk is negative (−0.54 percent). From the co-kurtosis result, the authors conclude that CAT bonds are the asset that raises the probability of extreme co-variations with the benchmark portfolio the least.

The authors also report results that temper these findings. The GARCH parameters of the marginal models are almost always jointly insignificant, and the copula's DCC parameter *a* is estimated at zero, which they note is inconsistent with their test of dynamic correlation. Their peer-performance test gives a 98 percent probability that the fund performs equally to the stock indices, and their most-diversified portfolio gives the fund a zero weight. They read their own beta cautiously: there is "a slight positive beta co-variance", so adding the fund to the benchmark portfolio "will not reduce its volatility" (pp. 304-305). They nevertheless cite Carayannopoulos and Perez (2015) in support (p. 298) and conclude that their results "confirm those of previous studies" (p. 307).

2.6 Other studies: divergent and supporting evidence

Two studies of CAT bond prices take the opposite side. Gürtler et al. (2016) find that the financial crisis and Hurricane Katrina both had a significant effect on secondary-market premiums, and that premiums are positively related to corporate bond spreads, a link that grew stronger during the crisis; they conclude that CAT bonds should not be regarded as zero-beta securities. Braun (2016) finds that the spreads of comparably rated corporate bonds help explain CAT bond spreads at issuance. Both show that the price investors demand for catastrophe risk moves with the price of credit risk.

A recent study supports the favourable view for a later crisis. Applying a DCC-GARCH model to an international portfolio of stocks, private equity, infrastructure and commodities before and during the COVID-19 pandemic, Feng and Zeng (2025) find that CAT bonds were a diversifier over the whole sample and a safe haven against stocks and private equity after the outbreak. The results hold under alternative specifications, including desmoothed returns. Two findings point the other way: hedge ratios are sensitive to extreme weather, and the correlation with commodities became stronger in falling markets during the pandemic. Table 1 summarises the three main articles.

Table 1. The three articles compared

	Carayannopoulos and Perez (2015)	**Drobetz, Schröder and Tegtmeier (2020)**	**Haffar and Le Fur (2022)**
Data	Swiss Re BB price index vs S&P 500, corporate and government bonds; 614 weekly returns, 2002-2013	Swiss Re Global Unhedged index vs six asset classes; 885 weekly returns, 2002-2018	GAMSTAR CAT bond fund vs six stock indices; 100 monthly returns, Nov. 2011 to Feb. 2020
Method	Diagonal-VEC MGARCH, Markov switching, dynamic hedge ratios	DCC-GARCH, then regression of the DCC on 10/5/1 percent quantile dummies	Student-t copula with asymmetric DCC; performance, VaR, co-moments
Stress	NBER crisis, Dec. 2007 to May 2009	Extreme weeks of the other asset; before vs after 2009	Co-kurtosis; no crisis in the sample

	Carayannopoulos and Perez (2015)	Drobetz, Schröder and Tegtmeier (2020)	Haffar and Le Fur (2022)
Key results	Correlation with S&P 500 before / during / after: 0.016 / 0.252 / 0.058; beta peaks at 0.014	Average correlation with stocks 0.24 before 2009, 0.10 after; post-crisis $m_2 = -0.091$	Beta 0.031; beta co-kurtosis 0.015; modified Sharpe ratio 0.112
Zero beta	Rejected in the crisis, holds outside it	Rejected on average; safe haven against stocks after 2009	Closest to zero beta (not tested formally)

Source: compiled by the author from the three articles.

2.7 What the literature agrees on, and what it leaves open

The studies agree that CAT bonds diversify stock portfolios and have low volatility and small betas. Their contradictions are of two kinds. The first concerns average behaviour: outside crises, Carayannopoulos and Perez cannot distinguish the correlation with the stock market from zero, a weak hedge, while Drobetz et al. find a positive and significant average correlation, a diversifier but not a hedge, and Gürtler et al. and Braun find that premiums move with corporate spreads. The second and sharper contradiction concerns stress: CAT bonds moved with markets after Lehman's failure and in extreme weeks before 2009, but Drobetz et al. find a safe haven after 2009, Feng and Zeng after the COVID-19 outbreak, and Haffar and Le Fur the weakest extreme co-movement of all the assets they study. Haffar and Le Fur see no contradiction because their question is diversification, not zero beta.

What the literature leaves open is why the answer changes. No study separates a loss on collateral from a fall in price driven by liquidity, and none conditions on the type of stress: NBER dates, extreme weeks of the other asset and a pre/post-outbreak split all treat a funding crisis and an equity sell-off alike. Section 3 addresses that gap.

3. Analysis

3.1 How far the evidence of each article goes

Each article answers its own question; none answers the question of this paper on its own. The strength of Carayannopoulos and Perez is that they treat beta as a variable and date the change: the correlation stayed near zero through the first nine months of the recession and jumped only when Lehman's failure hit the swap structures. Their limit is that a beta estimated on returns cannot say which part of the bond moved: a fall in the value of Lehman-guaranteed collateral and a rise in the premium demanded by investors who needed cash both appear as a positive beta, and their own account of 2008 describes both. Their treatment of Hurricane Ike is an argument by comparison, not a formal separation of the two shocks.

Drobetz et al. come closest to my question, but three features limit their safe-haven result. It rests on one coefficient out of three, and the arithmetic of their own model gives a weaker picture: because the dummies are nested, the implied correlation in the worst 5 percent of post-crisis weeks is about 0.05, or 0.01 if the insignificant m_1 is set to zero, and in the worst 1 percent of

weeks the point estimates add up to about 0.15, above the 0.10 of ordinary weeks.[1] This is at most a weak safe haven in Baur and McDermott's terms, not a strong one. Second, the pre-crisis period ends in December 2008, so it contains the months after Lehman's failure. Third, the dummies identify stress by the other asset's return only, so they cannot tell an equity sell-off from a funding crisis.

Haffar and Le Fur's favourable result has the narrowest base. It rests on co-kurtosis betas rather than a copula tail-dependence coefficient, the copula adds little since its correlation dynamics are not identified, and the performance advantage is not statistically distinct (Section 2.5). Ranking first among six stock indices that are strongly correlated with each other and with the equity benchmark is a modest test. The fund series also mixes CAT bond risk with a manager's choices. Above all, the sample ends days before the March 2020 sell-off and contains no funding crisis and no collateral failure.

The disagreement is therefore smaller than the labels suggest: with betas of 0.014 at the 2010 peak and 0.031 in calm years, the studies disagree mainly on whether a small exposure contradicts the zero-beta claim, and on whether 2008, which a simple account would call a special case, says anything about later crises. My argument is that the bonds changed, the crises differed and the methods differed, and that a decomposition separates the first two and makes a prediction the simple account does not (Section 3.6).

3.2 My contribution: a decomposition of the CAT bond return

I write the return of a CAT bond, or of an index of CAT bonds, over one period as

$$R_t = r_t + s - \ell_t - D \cdot \Delta\pi_t - k_t \tag{1}$$

where r_t is the return on the collateral, s is the contractual spread accrued over the period, ℓ_t is the loss caused by catastrophes (a realised loss, or a revaluation of the expected loss after an event), $\Delta\pi_t$ is the change in the premium π that investors require above the expected loss, D is the bond's spread duration, and k_t is the collateral loss caused by a counterparty failure or a fall in the value of the collateral assets. Defining π net of the expected loss keeps a revision of the expected loss in ℓ_t only. The insurance core is $s - \ell_t$ and the financial wrapper is $r_t - D \cdot \Delta\pi_t - k_t$.

The beta of the bond with respect to the market return R_m is then the sum of the betas of its components, $\beta_x = \mathrm{Cov}(x_t, R_{m,t})/\mathrm{Var}(R_{m,t})$, where β_π is the beta of $\Delta\pi_t$:[2]

$$\beta = \beta_r - \beta_\ell - D \cdot \beta_\pi - \beta_k \tag{2}$$

Three assumptions, stated so that they can be challenged, turn equation (2) into a usable result:

- **A1.** For natural perils, catastrophe losses do not depend on financial markets: $\beta_\ell = 0$. This is the zero-beta argument itself.

[1] A week in the worst 5 percent of stock returns is also in the worst 10 percent but not in the worst 1 percent, so $d_{10\%} = d_{5\%} = 1$ and $d_{1\%} = 0$, and the fitted correlation is $m_0 + m_1 + m_2 \approx 0.10 + 0.04 - 0.09 = 0.05$. Setting the insignificant m_1 to zero gives $0.10 - 0.09 = 0.01$. In the worst 1 percent of weeks all three dummies equal one: $0.10 + 0.04 - 0.09 + 0.10 = 0.15$.

[2] Covariance is linear, $\mathrm{Cov}(a + b, R_m) = \mathrm{Cov}(a, R_m) + \mathrm{Cov}(b, R_m)$, and dividing by $\mathrm{Var}(R_m)$ gives the same rule for betas. D is treated as constant within a period, so the beta of $D \cdot \Delta\pi_t$ is $D \cdot \beta_\pi$; the fixed spread s has a beta of zero.

- **A2.** The collateral earns a short-term money-market rate whose weekly variation is negligible: $\beta_r \approx 0$.
- **A3.** Within a week, D is constant and the price responds linearly to $\Delta\pi$, a first-order approximation for bonds with one to two years of remaining life.

Under A1 and A2, what is left is

$$\beta \approx -D \cdot \beta_\pi - \beta_k \tag{3}$$

Under these assumptions, the measured beta of a natural-catastrophe bond is the beta of its financial wrapper. I call the first term the premium channel and the second the collateral channel. When markets fall, the required premium rises ($\beta_\pi < 0$) and collateral losses become more likely ($\beta_k < 0$), so both terms push the beta above zero. The channels differ in cost: a premium-driven fall is recovered by maturity if no catastrophe occurs, so it hurts mainly a holder who must sell, while a collateral loss is permanent.

A1 holds only approximately: the short high-correlation regimes after Katrina, the 2011 Japanese earthquake and Sandy make β_ℓ small rather than exactly zero, and π is itself the price of catastrophe risk. What the decomposition separates is the occurrence of catastrophes, which markets do not drive, from the price and the packaging of catastrophe risk, which they can.

Equation (3) leads to three propositions.

1. **P1.** In calm markets the collateral channel is closed and the premium moves little, so β is small. It need not be zero, since the link between CAT premiums and corporate spreads (Gürtler et al., 2016; Braun, 2016) exists outside crises, consistent with the post-2009 average correlation of about 0.10 (Drobetz et al., 2020).
2. **P2.** The collateral channel can open only when the collateral is exposed to the credit of a financial institution: it could before 2009, and it is close to closed for deals whose collateral is held in Treasury funds or government-backed notes.
3. **P3.** The premium channel opens when holders need cash as the market falls: they sell, and dealers short of capital buy only at a higher premium. Market and funding liquidity then reinforce each other (Brunnermeier and Pedersen, 2009), and slow-moving capital pushes prices below fundamentals before they recover (Duffie, 2010). If the capital available to bear catastrophe risk shrinks with financial wealth, its price also rises with credit spreads, which is what Gürtler et al. and Braun observe.

The zero-beta claim fails at its root when the peril is itself a macroeconomic shock ($\beta_\ell \neq 0$): pandemic bonds, and in my view systemic cyber bonds, belong to a different asset class.

3.3 Four stress regimes

Table 2 classifies stress episodes by whether the collateral was exposed to bank credit and whether investors needed cash (a funding crisis) or not (an equity sell-off).

Table 2. Stress regimes implied by the decomposition, with the evidence for each

	Funding crisis (investors need cash)	**Equity sell-off, no forced selling**
Collateral exposed to bank credit (swap structures issued before 2009)	Both channels open: beta turns positive. *Evidence:* correlation with stocks jumps after September 2008 (Carayannopoulos and Perez, 2015) and rises in extreme weeks before 2009 (Drobetz et al., 2020).	Only the insurance core is tested: beta close to zero. *Evidence:* correlation near zero in the first nine months of the recession (Carayannopoulos and Perez, 2015).
Collateral in Treasury funds or government-backed notes (deals issued after 2009)	Premium channel only: a small fall, later reversed; at most a weak safe haven. *Evidence:* safe haven after the COVID-19 outbreak (Feng and Zeng, 2025); fall of 0.81 percent in March 2020 (Insurance Insider ILS, 2020).	Neither channel open: beta close to zero, weak safe haven. *Evidence:* implied correlation of 0.01 to 0.05 in the worst post-2009 stock weeks (Drobetz et al., 2020); beta of 0.031 (Haffar and Le Fur, 2022).

Source: classification by the author, based on the studies cited in each cell.

Each main result fits one cell. Since the grid was built after reading the evidence, this shows compatibility, not proof.

The top-left cell contains one episode, the months after Lehman's failure. The timing fits the collateral channel but cannot isolate it, since no forced selling is documented in the first nine months of the recession. The positive pre-2009 extreme-week coefficients of Drobetz et al. belong here, since their pre-crisis period ends in December 2008, and the persistence of the regime until early 2011 fits swap-backed bonds that stayed outstanding and capital that returned slowly (Duffie, 2010).

The bottom-right cell explains the favourable results. As swap-backed deals matured after 2009 the collateral channel closed, and I assume, as Table 3 can check, that post-crisis stock declines brought no forced selling on the scale of late 2008. The grid then predicts a beta close to zero and a weak safe haven, as observed. The whole sample of Haffar and Le Fur lies in this cell, which is why their result is favourable and says nothing about funding crises.

March 2020 is the main test of the bottom-left cell: safe collateral combined with a severe scramble for cash. The framework predicts a fall driven by forced sales, no collateral loss, and a recovery. The Swiss Re price return index fell by 0.81 percent in the week to 20 March, with markdowns linked to bulk lists of bonds put up for sale (Insurance Insider ILS, 2020), and by May the sell-off was described as temporary (The Insurer, 2020). No natural catastrophe drove the fall and the collateral was safe, so in equation (1) $\ell_t = k_t = 0$ and the fall equals $D \cdot \Delta\pi_t$. With a spread duration of one to two years, the required premium rose by $0.81/D$, about 40 to 80 basis points,[3] of the same order as the liquidity premium estimated by Herrmann and Hibbeln (2023), 98 basis points on average and 141 for high-risk bonds. The 0.81 percent is a reported index move, not a published estimate, so the result is an order of magnitude, made credible by its agreement with this peer-reviewed measure. The premium channel thus opened but stayed small,

[3] The floating part of the coupon resets and barely moves the price; the fixed spread does not reset. If investors require $\Delta\pi$ more per year, the bond pays $\Delta\pi$ too little in each remaining year, so its price falls by about $D \cdot \Delta\pi$, with D close to the remaining life. With maturities of three to four years at issue, the average bond in an index has about 1.5 to 2 years left; one to two years allows for bonds near maturity. A price index excludes coupons, so r_t and s drop out. Hence $\Delta\pi \approx 0.81/2 \approx 0.40$ percent (40 basis points) for $D = 2$ and $0.81/1 = 0.81$ percent (about 80 basis points) for $D = 1$.

consistent with Feng and Zeng still finding a safe haven; what the simple account does not predict is a fall linked to forced sales and then reversed.

3.4 Measurement choices that move the answer

Part of the divergence is statistical. The first source is the gap between economic and statistical significance: CAT bond returns are much less volatile than stock returns, so a significant correlation can go with a tiny beta. In the data of Carayannopoulos and Perez, the crisis correlation of 0.25 goes with a beta that peaked at 0.014: a 20 percent fall in the S&P 500 would have cost a CAT bond investor about 0.3 percent.[4] The zero-beta claim is rejected statistically and close to true economically, and the articles differ partly in which of the two they stress.

The second is conditioning on volatility. Suppose the CAT bond return is β times the market return plus an independent shock with volatility σ_ε. The correlation is then $\rho = \beta \cdot \sigma_m / \sqrt{(\beta^2 \sigma_m^2 + \sigma_\varepsilon^2)}$, which rises with the market's volatility σ_m whenever $\beta > 0$ (Forbes and Rigobon, 2002). In the data of Carayannopoulos and Perez, the weekly volatility of the S&P 500 rose from 1.91 to 4.49 percent, which at a constant beta and idiosyncratic volatility would lift the correlation only from 0.016 to under 0.04.[5] The rest of the rise to 0.252 is not mechanical, and their beta, which is free of this bias, also rose.

The third, smoothing, matters less than theory suggests. A smoothed index biases beta and correlation toward zero (Getmansky et al., 2004), but the results of Drobetz et al. and of Feng and Zeng hold with desmoothed returns. Smoothing also produces positive autocorrelation, while the index of Drobetz et al. has negative autocorrelation, which points instead to noise in quoted prices that reverses the following week, a problem desmoothing does not correct.

The fourth, the choice of proxy, measure and window, matters most. Before the crisis the correlation with stocks is 0.016 in Carayannopoulos and Perez but about 0.24 in Drobetz et al. The few weeks of late 2008 in the second sample cannot explain the gap,[6] which must come from the series (BB price index or global total return index), the measure (sample correlation or mean DCC) and the comparison index.

3.5 Rival explanations

The decomposition is not the only way to read the evidence. Three rival explanations can be tested against what the articles report.

The first is that Hurricane Ike, not the financial crisis, caused the 2008 correlation. A catastrophe shock should raise the correlation briefly and then let it fade once losses are estimated, as after

[4] By definition $\beta = \rho \cdot \sigma_{CAT}/\sigma_m$. In the crisis sample $\sigma_{CAT}/\sigma_m = 0.41/4.49 \approx 0.09$, so a correlation of 0.25 implies a beta of about $0.25 \times 0.09 \approx 0.02$, close to the dynamic peak of 0.014. With $\beta = 0.014$, a 20 percent fall in the S&P 500 moves the CAT bond by $0.014 \times 20 = 0.28$ percent.

[5] Before the crisis, $\rho = 0.016$, $\sigma_{CAT} = 0.37$ and $\sigma_m = 1.91$ percent. Since β is tiny, practically all CAT bond volatility is idiosyncratic ($\sigma_\varepsilon \approx 0.37$), and $\beta \approx 0.016 \times 0.37/1.91 \approx 0.003$. Keeping β and σ_ε fixed and setting $\sigma_m = 4.49$ gives $\rho \approx (0.003 \times 4.49)/\sqrt{((0.003 \times 4.49)^2 + 0.37^2)} \approx 0.0139/0.370 \approx 0.038$.

[6] From mid-September to December 2008 is about 15 of the roughly 365 weekly observations of 2002-2008. Even if the correlation had been 1 in all 15 weeks and 0.016 in the other 350, its average would be $(15 \times 1 + 350 \times 0.016)/365 \approx 0.06$, far below 0.24.

Katrina, the 2011 Japanese earthquake and Sandy. In 2008 the high-correlation regime with stocks lasted about 30 months, and the correlation rose with corporate bonds while it fell with government bonds (Section 2.3). A hurricane explains neither the duration nor the link with credit. Ike may have triggered the move, as Carayannopoulos and Perez argue, but it cannot have sustained it.

The second is that CAT bonds look safer after 2009 only because markets were calmer. This is the simple account in another form, and the evidence does not fully reject it. Two points favour the decomposition: the post-2009 sample of Drobetz et al. contains extreme stock weeks, and in March 2020 the fall stayed below 1 percent and reversed despite a severe scramble for cash. Both rest on few weeks, hence test 1 of Table 3.

The third is that the measurement changed rather than the bonds. Quote noise lowers measured correlations, but in every period, so it cannot explain a change between periods, and the desmoothed estimates of Drobetz et al. and Feng and Zeng do not move.

One implication follows that none of the articles draws, and I state it as a fourth proposition. **P4.** The premium channel, and with it the crisis beta, rises with the share of bonds held by investors who must meet redemptions at short notice. If that share grows, the channel that stayed small in March 2020 can widen in the next funding crisis. The favourable post-2009 evidence thus describes a market with a given investor base, not a permanent property of the asset.

3.6 Proposed solutions

Tests that could prove the framework wrong. Table 3 sets out four tests, their data and what each account predicts.

Table 3. Tests that would discriminate between the decomposition and the simple account

Test	Data and measures	Decomposition predicts	Simple account predicts
1. $\mathbf{DCC_t = c_0 + c_1 \cdot Ext_t + c_2 \cdot Fund_t + c_3 \cdot Ext_t \cdot Fund_t}$	Weekly DCC of a Swiss Re index with stocks; Ext_t: worst 5 percent of stock weeks; $Fund_t$: top decile of the commercial paper–Treasury bill spread	$c_1 \approx 0$; $c_3 > 0$ before and after 2009, smaller after (P2, P3)	$c_3 > 0$ before 2009 only
2. Split of the 2008 beta	Prices of Lehman-swap bonds and of other CAT bonds, September to December 2008	A gap between groups (collateral channel) and a common fall rising with duration (premium channel)	No prediction on the split
3. Path after a funding shock	Index prices and bid lists around March 2020	A fall linked to bond sales, no collateral loss, recovery within months	Little or no fall in a safe-haven asset
4. Investor base	Holdings by funds with frequent redemptions, across funding-stress weeks	Larger stress betas when that share is higher (P4)	No relation

Source: proposed by the author.

A fifth test would re-estimate Haffar and Le Fur's copula measures on weekly data including 2008 and March 2020, where the decomposition predicts stronger extreme co-movement in funding-crisis weeks. In every test, catastrophe weeks such as Ike's need their own dummies,

betas should be reported next to correlations, crisis correlations adjusted for volatility (Forbes and Rigobon, 2002), and TRACE prices used where possible (Herrmann and Hibbeln, 2023).

Portfolio and market rules that remove what can be removed.

- ***Keep the collateral channel closed:*** require collateral in Treasury funds or government-backed notes, held in trust and marked to market, and check each issue.
- ***Contain the premium channel:*** selling in a crisis turns a temporary markdown into a realised loss, so funds with frequent redemptions need notice periods or cash buffers (P4), and value-at-risk should be stressed on a funding crisis such as March 2020.
- ***Classify perils honestly:*** bonds on macroeconomic perils, such as pandemics, are not zero-beta exposures.

3.7 Validity of the analysis

Internal validity. The grid of Table 2 was built after reading the evidence, so its fit shows compatibility, not confirmation; only the tests of Table 3, on data not used to build it, can confirm it. Two cells are thin, for different reasons. The top-left can only contain 2008, because the collateral rules adopted afterwards (Section 2.1) removed the structure it describes; this is a feature of the market, not a gap in the review. The bottom-left rests mainly on March 2020 because funding crises are rare, which is why test 1 uses every week of funding stress rather than named episodes. The magnitudes depend on assumption A3 and on reported market falls, so the 40 to 80 basis points are an order of magnitude, not an estimate.

External validity. The evidence comes from two Swiss Re indices built from dealer quotes and from one fund. Individual bonds, other perils and other investor bases may behave differently, and the conclusions do not extend to pandemic or systemic cyber bonds, for which A1 fails.

3.8 Conclusion of the analysis

Whether CAT bonds are zero-beta assets depends on which part of the bond the question refers to. For natural perils the insurance core is zero-beta; the traded bond has a small beta in calm markets, one that rose, though it stayed small, in 2008, and one that can rise in a funding crisis. The safe-haven property depends on the asset's structure, the type of stress and the investor base, not only on the state of the market.

4. Conclusion

This paper asked to what extent CAT bonds retain their diversification benefits during financial stress, and why studies disagree. The three main articles sample different cells of Table 2 with different proxies and windows, so their conclusions can diverge while their estimates are compatible.

The answer to the research question is therefore conditional. CAT bonds kept their diversification benefit in every episode reviewed; even after Lehman's failure their beta peaked at 0.014. Their safe-haven property holds, weakly, when the collateral is safe and holders are not forced to sell,

and should erode when either condition fails. Investors should treat them as a diversifier, not as crisis insurance.

The main limit is that the framework has not been tested: it rests on published results, and Section 3.7 sets out what this implies for its validity. Running the tests of Table 3 is the natural next step.

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